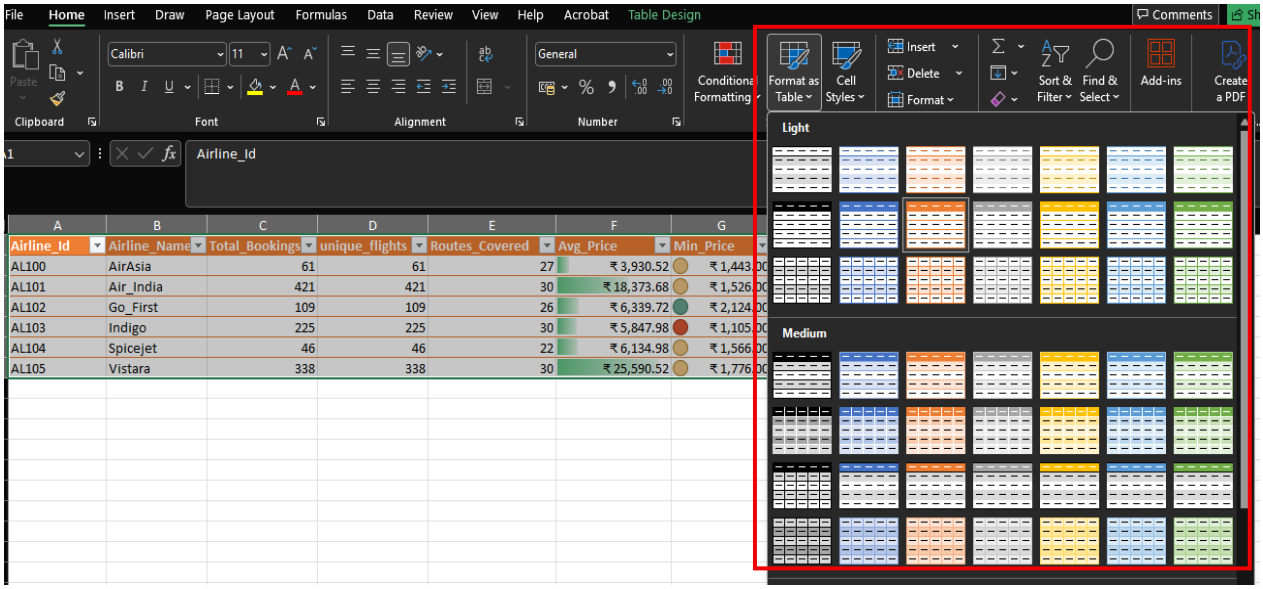
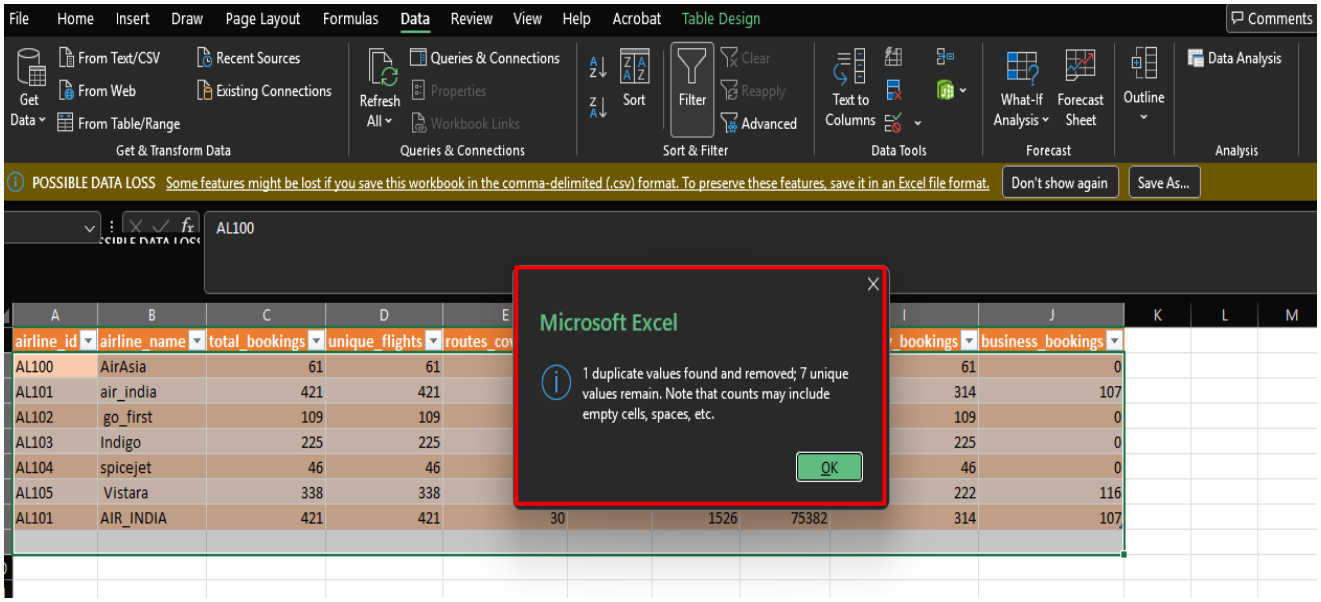
# DATA PRE-PROCESSING:

## Converted Raw data into Tables




### 1. AIRLINES

- Checked the Airlines table for duplicate values.
- Checked and handled missing values.
- Standardized the airline-related data.
- Created calculations for Total Bookings, Routes Covered, Average Price, Minimum Price and Maximum Price.
- Calculated Economy and Business class bookings.



POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

=TRIM(PROPER([@[airline_name]]))

| airline_id | airline_name | total_bookings | unique_flights | routes_covered | avg_price | min_price | max_price | economy_bookings | business_bookings | Names |
|---|---|---|---|---|---|---|---|---|---|---|
| AL100 | AirAsia | 61 | 61 | 27 | 3,465 | 1443 | 7484 | 61 | 0 | Airasia |
| AL101 | air_india | 421 | 421 | 30 | â‚¹19,084 | 1526 | 75382 | 314 | 107 | Air_India |
| AL102 | go_first | 109 | 109 | 26 | 6,255 | 2124 | - | 109 | 0 | Go_First |
| AL103 | Indigo | 225 | 225 | 30 | NA | | 24561 | 225 | 0 | Indigo |
| AL104 | spicejet | 46 | 46 | 22 | â‚¹6,045 | 1566 | N/A | 46 | 0 | Spicejet |
| AL105 | Vistara | 338 | 338 | 30 | 26599 | 1776 | 110936 | 222 | 116 | Vistara |
| AL101 | AIR_INDIA | 421 | 421 | 30 | | 1526 | 75382 | 314 | 107 | Air_India |

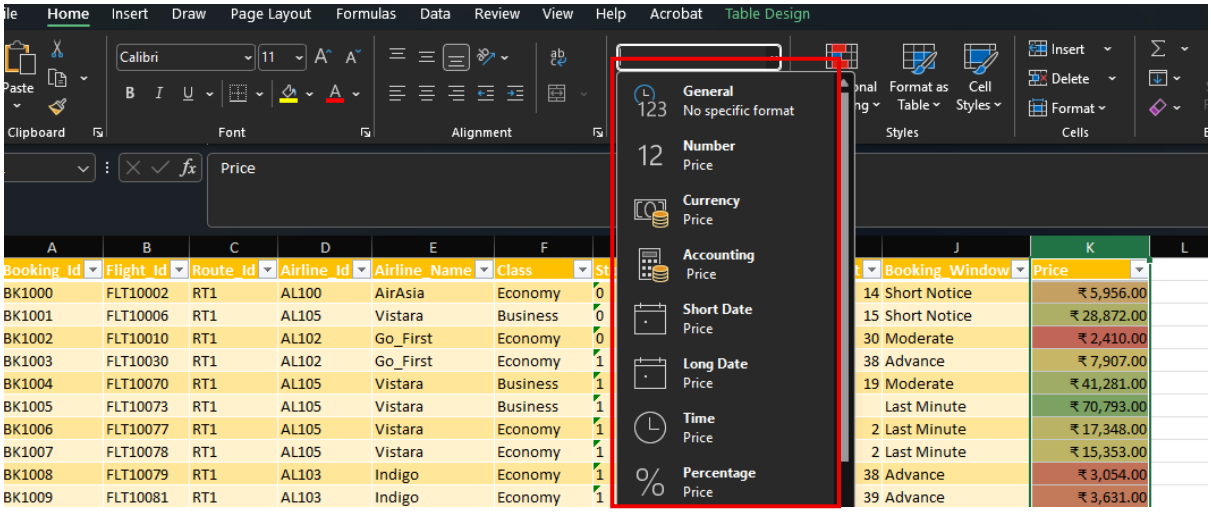
| Airline_Id | Airline_Name | Total_Bookings | unique_flights | Routes_Covered | Avg_Price | Min_Price | Max_Price | Economy_Bookings | Business_Bookings |
|---|---|---|---|---|---|---|---|---|---|
| AL100 | AirAsia | 61 | 61 | 27 | ₹3,930.52 | ₹1,443.00 | ₹7,484.00 | 61 | 0 |
| AL101 | Air_India | 421 | 421 | 30 | ₹18,373.68 | ₹1,526.00 | ₹75,382.00 | 314 | 107 |
| AL102 | Go_First | 109 | 109 | 26 | ₹6,339.72 | ₹2,124.00 | ₹31,773.00 | 109 | 0 |
| AL103 | Indigo | 225 | 225 | 30 | ₹5,847.98 | ₹1,105.00 | ₹24,561.00 | 225 | 0 |
| AL104 | Spicejet | 46 | 46 | 22 | ₹6,134.98 | ₹1,566.00 | ₹27,769.00 | 46 | 0 |
| AL105 | Vistara | 338 | 338 | 30 | ₹25,590.52 | ₹1,776.00 | ₹1,10,936.00 | 222 | 116 |

## 2. BOOKINGS

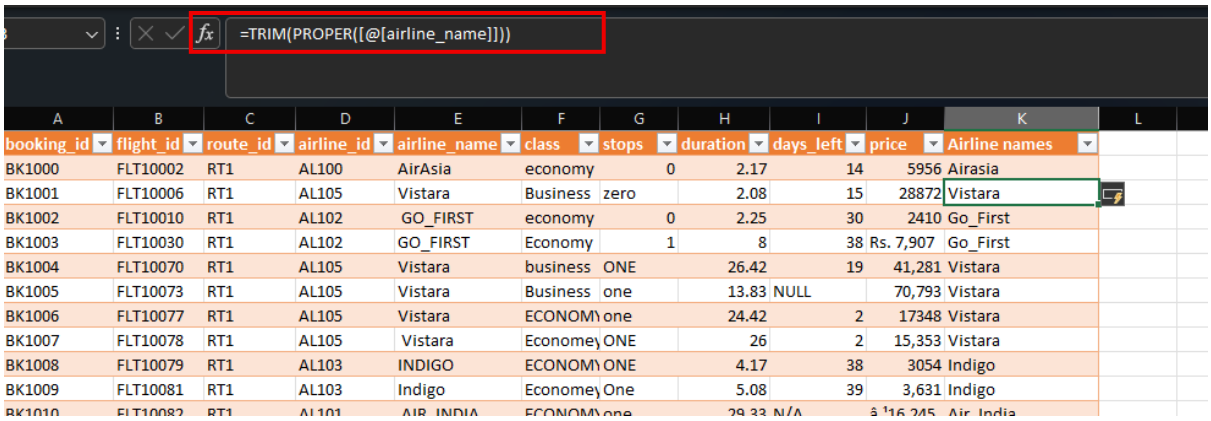
- Checked the Bookings table for duplicate values.
- Checked and handled missing values.
- Corrected the data types of the required columns.
- Used lookup functions to retrieve the required airline and flight information.
- Created a new **Booking Window** column using Days Left.
- Categorized bookings into Last Minute, Short Notice, Moderate and Advance.

=IF(I4<=7,"Last Minute",IF(I4<=15,"Short Notice",IF(I4<=30,"Moderate",IF(I4>=31,"Advance"))))

| Booking_Id | Flight_Id | Route_Id | Airline_Id | Airline_Name | Class | Stops | Duration | Days_Left | Booking_Window | Price |
|---|---|---|---|---|---|---|---|---|---|---|
| BK1000 | FLT10002 | RT1 | AL100 | AirAsia | Economy | 0 | 2.17 | 14 | Short Notice | ₹5,956.00 |
| BK1001 | FLT10006 | RT1 | AL105 | Vistara | Business | 0 | 2.08 | 15 | Short Notice | ₹28,872.00 |
| BK1002 | FLT10010 | RT1 | AL102 | Go_First | Economy | 0 | 2.25 | 30 | Moderate | ₹2,410.00 |
| BK1003 | FLT10030 | RT1 | AL102 | Go_First | Economy | 1 | 8 | 38 | Advance | ₹7,907.00 |
| BK1004 | FLT10070 | RT1 | AL105 | Vistara | Business | 1 | 26.42 | 19 | Moderate | ₹41,281.00 |
| BK1005 | FLT10073 | RT1 | AL105 | Vistara | Business | 1 | 13.83 | | Last Minute | ₹70,793.00 |
| BK1006 | FLT10077 | RT1 | AL105 | Vistara | Economy | 1 | 24.42 | 2 | Last Minute | ₹17,348.00 |
| BK1007 | FLT10078 | RT1 | AL105 | Vistara | Economy | 1 | 26 | 2 | Last Minute | ₹15,353.00 |
| BK1008 | FLT10079 | RT1 | AL103 | Indigo | Economy | 1 | 4.17 | 38 | Advance | ₹3,054.00 |
| BK1009 | FLT10081 | RT1 | AL103 | Indigo | Economy | 1 | 5.08 | 39 | Advance | ₹3,631.00 |
| BK1010 | FLT10082 | RT1 | AL101 | Air_India | Economy | 1 | 29.33 | | Last Minute | ₹16,245.00 |
| BK1011 | FLT10083 | RT1 | AL105 | Vistara | Economy | 1 | 4.42 | 46 | Advance | ₹4,381.00 |
| BK1012 | FLT10084 | RT1 | AL105 | Vistara | Economy | 1 | 17 | 22 | Moderate | ₹7,575.00 |



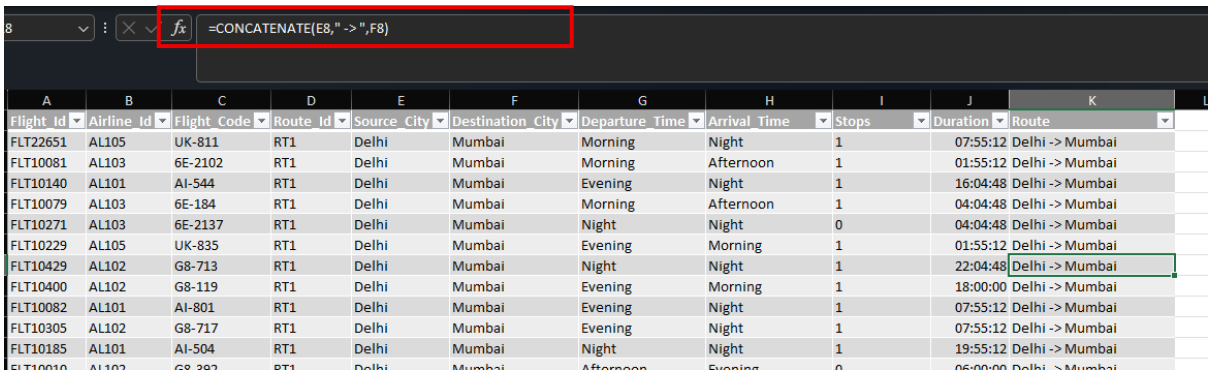

| Booking_Id | Flight_Id | Route_Id | Airline_Id | Airline_Name | Class | St | Booking_Window | Price |
|---|---|---|---|---|---|---|---|---|
| BK1000 | FLT10002 | RT1 | AL100 | AirAsia | Economy | 0 | 14 Short Notice | ₹5,956.00 |
| BK1001 | FLT10006 | RT1 | AL105 | Vistara | Business | 0 | 15 Short Notice | ₹28,872.00 |
| BK1002 | FLT10010 | RT1 | AL102 | Go_First | Economy | 0 | 30 Moderate | ₹2,410.00 |
| BK1003 | FLT10030 | RT1 | AL102 | Go_First | Economy | 1 | 38 Advance | ₹7,907.00 |
| BK1004 | FLT10070 | RT1 | AL105 | Vistara | Business | 1 | 19 Moderate | ₹41,281.00 |
| BK1005 | FLT10073 | RT1 | AL105 | Vistara | Business | 1 | Last Minute | ₹70,793.00 |
| BK1006 | FLT10077 | RT1 | AL105 | Vistara | Economy | 1 | 2 Last Minute | ₹17,348.00 |
| BK1007 | FLT10078 | RT1 | AL105 | Vistara | Economy | 1 | 2 Last Minute | ₹15,353.00 |
| BK1008 | FLT10079 | RT1 | AL103 | Indigo | Economy | 1 | 38 Advance | ₹3,054.00 |
| BK1009 | FLT10081 | RT1 | AL103 | Indigo | Economy | 1 | 39 Advance | ₹3,631.00 |



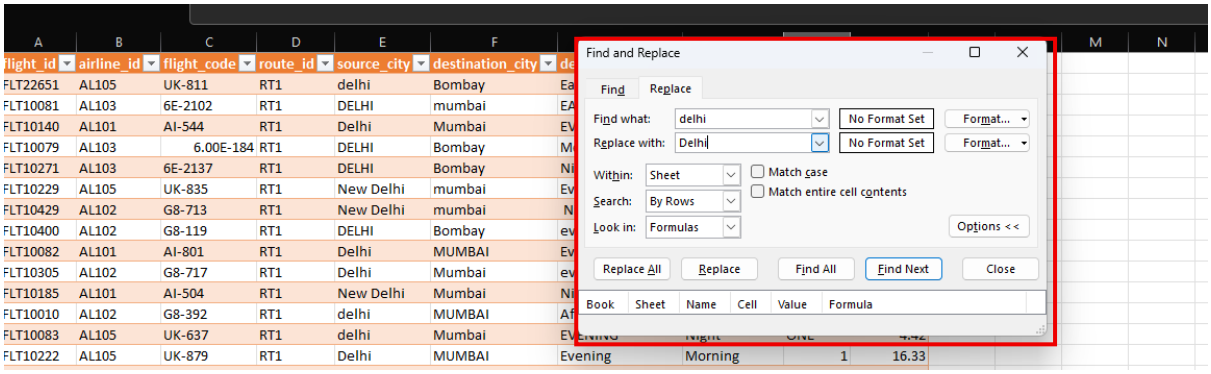

| booking_id | flight_id | route_id | airline_id | airline_name | class | stops | duration | days_left | price | Airline names |
|---|---|---|---|---|---|---|---|---|---|---|
| BK1000 | FLT10002 | RT1 | AL100 | AirAsia | economy | 0 | 2.17 | 14 | 5956 | Airasia |
| BK1001 | FLT10006 | RT1 | AL105 | Vistara | Business | zero | 2.08 | 15 | 28872 | Vistara |
| BK1002 | FLT10010 | RT1 | AL102 | GO_FIRST | economy | 0 | 2.25 | 30 | 2410 | Go_First |
| BK1003 | FLT10030 | RT1 | AL102 | GO_FIRST | Economy | 1 | 8 | 38 | Rs. 7,907 | Go_First |
| BK1004 | FLT10070 | RT1 | AL105 | Vistara | business | ONE | 26.42 | 19 | 41,281 | Vistara |
| BK1005 | FLT10073 | RT1 | AL105 | Vistara | Business | one | 13.83 | NULL | 70,793 | Vistara |
| BK1006 | FLT10077 | RT1 | AL105 | Vistara | ECONOM\ | one | 24.42 | 2 | 17348 | Vistara |
| BK1007 | FLT10078 | RT1 | AL105 | Vistara | Economey | ONE | 26 | 2 | 15,353 | Vistara |
| BK1008 | FLT10079 | RT1 | AL103 | INDIGO | ECONOM\ | ONE | 4.17 | 38 | 3054 | Indigo |
| BK1009 | FLT10081 | RT1 | AL103 | Indigo | Economey | One | 5.08 | 39 | 3,631 | Indigo |
| BK1010 | FLT10082 | RT1 | AL101 | AIR_INDIA | ECONOM\ | one | 29.33 | N/A | â¹16,245 | Air_India |

## 3. FLIGHTS

- Checked the Flights table for duplicate values.
- Checked and handled missing values.
- Corrected data types of the required columns.
- Standardized Source City and Destination City values.
- Created a new **Route** column by combining Source City and Destination City.
- Checked and standardized flight-related information.




| Flight_Id | Airline_Id | Flight_Code | Route_Id | Source_City | Destination_City | Departure_Time | Arrival_Time | Stops | Duration | Route |
|---|---|---|---|---|---|---|---|---|---|---|
| FLT22651 | AL105 | UK-811 | RT1 | Delhi | Mumbai | Morning | Night | 1 | 07:55:12 | Delhi -> Mumbai |
| FLT10081 | AL103 | 6E-2102 | RT1 | Delhi | Mumbai | Morning | Afternoon | 1 | 01:55:12 | Delhi -> Mumbai |
| FLT10140 | AL101 | AI-544 | RT1 | Delhi | Mumbai | Evening | Night | 1 | 16:04:48 | Delhi -> Mumbai |
| FLT10079 | AL103 | 6E-184 | RT1 | Delhi | Mumbai | Morning | Afternoon | 1 | 04:04:48 | Delhi -> Mumbai |
| FLT10271 | AL103 | 6E-2137 | RT1 | Delhi | Mumbai | Night | Night | 0 | 04:04:48 | Delhi -> Mumbai |
| FLT10229 | AL105 | UK-835 | RT1 | Delhi | Mumbai | Evening | Morning | 1 | 01:55:12 | Delhi -> Mumbai |
| FLT10429 | AL102 | G8-713 | RT1 | Delhi | Mumbai | Night | Night | 1 | 22:04:48 | Delhi -> Mumbai |
| FLT10400 | AL102 | G8-119 | RT1 | Delhi | Mumbai | Evening | Morning | 1 | 18:00:00 | Delhi -> Mumbai |
| FLT10082 | AL101 | AI-801 | RT1 | Delhi | Mumbai | Evening | Night | 1 | 07:55:12 | Delhi -> Mumbai |
| FLT10305 | AL102 | G8-717 | RT1 | Delhi | Mumbai | Evening | Night | 1 | 07:55:12 | Delhi -> Mumbai |
| FLT10185 | AL101 | AI-504 | RT1 | Delhi | Mumbai | Night | Night | 1 | 19:55:12 | Delhi -> Mumbai |
| FLT10010 | AL102 | G8-382 | RT1 | Delhi | Mumbai | Afternoon | Evening | 0 | 06:00:00 | Delhi -> Mumbai |



| flight_id | airline_id | flight_code | route_id | source_city | destination_city | | | | |
|---|---|---|---|---|---|---|---|---|---|
| FLT22651 | AL105 | UK-811 | RT1 | delhi | Bombay | Ea | | | |
| FLT10081 | AL103 | 6E-2102 | RT1 | DELHI | mumbai | EA | | | |
| FLT10140 | AL101 | AI-544 | RT1 | Delhi | Mumbai | EV | | | |
| FLT10079 | AL103 | 6.00E-184 | RT1 | DELHI | Bombay | M | | | |
| FLT10271 | AL103 | 6E-2137 | RT1 | DELHI | Bombay | Ni | | | |
| FLT10229 | AL105 | UK-835 | RT1 | New Delhi | mumbai | Ev | | | |
| FLT10429 | AL102 | G8-713 | RT1 | New Delhi | mumbai | N | | | |
| FLT10400 | AL102 | G8-119 | RT1 | DELHI | Bombay | ev | | | |
| FLT10082 | AL101 | AI-801 | RT1 | Delhi | MUMBAI | Ev | | | |
| FLT10305 | AL102 | G8-717 | RT1 | Delhi | Mumbai | ev | | | |
| FLT10185 | AL101 | AI-504 | RT1 | New Delhi | Mumbai | Ni | | | |
| FLT10010 | AL102 | G8-392 | RT1 | delhi | MUMBAI | Af | | | |
| FLT10083 | AL105 | UK-637 | RT1 | delhi | Mumbai | Ev | | | |
| FLT10222 | AL105 | UK-879 | RT1 | Delhi | MUMBAI | Evening | Morning | 1 | 16.33 |



## 4. ROUTES

- Checked the Routes table for duplicate values.
- Checked and handled missing values.
- Calculated Total Bookings for each route.
- Calculated Unique Flights and Airlines Serving each route.
- Calculated Average Price, Minimum Price and Maximum Price.
- Calculated Average Duration.
- Calculated Economy and Business bookings.
- Used COUNTIF, COUNTIFS, AVERAGEIF, AVERAGEIFS, MINIFS and MAXIFS functions.

D4 =COUNTIF(Flights!D:D,A4)

| Route_Id | Source_City | Destination_City | Total_Bookings | Unique_Flights | Airlines_Serving | Avg_Price | Min_Price | Max_Price | Avg_Duration_Hrs | Economy_Bookings | Business_ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RT1 | Delhi | Mumbai | 40 | 40 | 5 | ₹15,969 | ₹2,381 | ₹71,056 | 10.91 | 30 | |
| RT2 | Delhi | Bengaluru | 40 | 40 | 6 | ₹13,093 | ₹3,090 | ₹54,777 | 11.64 | 34 | |
| RT3 | Delhi | Kolkata | 40 | 40 | 6 | ₹13,357 | ₹2,482 | ₹53,209 | 14.38 | 34 | |
| RT4 | Delhi | Hyderabad | 40 | 40 | 6 | ₹12,376 | ₹2,050 | ₹61,889 | 12.65 | 33 | |
| RT5 | Delhi | Chennai | 40 | 40 | 6 | ₹19,603 | ₹1,998 | ₹70,323 | 15.80 | 29 | |
| RT6 | Mumbai | Delhi | 40 | 40 | 6 | ₹14,809 | ₹2,336 | ₹59,964 | 10.55 | 32 | |
| RT7 | Mumbai | Bengaluru | 40 | 40 | 5 | ₹12,375 | ₹2,074 | ₹74,880 | 13.41 | 37 | |
| RT8 | Mumbai | Kolkata | 40 | 40 | 5 | ₹12,740 | ₹2,835 | ₹64,285 | 14.78 | 36 | |
| RT9 | Mumbai | Hyderabad | 40 | 40 | 5 | ₹21,093 | ₹2,111 | ₹60,365 | 16.16 | 26 | |
| RT10 | Mumbai | Chennai | 40 | 40 | 6 | ₹17,123 | ₹1,890 | ₹69,885 | 15.98 | 33 | |
| RT11 | Bengaluru | Delhi | 40 | 40 | 6 | ₹11,996 | ₹2,723 | ₹66,411 | 11.72 | 34 | |
| RT12 | Bengaluru | Mumbai | 40 | 40 | 5 | ₹19,003 | ₹3,012 | ₹83,916 | 13.83 | 31 | |

```
{=COUNTA(UNIQUE(FILTER(Flights!A:A,(Flights!E:E=B2)*(Flights!F:F=C2))))}
```

| Route_Id | Source_City | Destination_City | Total_Bookings | Unique_Flights | Airlines_Serving | Avg_Price | Min_Price | Max_Price | Avg_Duration_Hrs | Economy_Bookings |
|---|---|---|---|---|---|---|---|---|---|---|
| RT1 | Delhi | Mumbai | 40 | 40 | 5 | ₹15,969 | ₹2,381 | ₹71,056 | 10.91 | 30 |
| RT2 | Delhi | Bengaluru | 40 | 40 | 6 | ₹13,093 | ₹3,090 | ₹54,777 | 11.64 | 34 |
| RT3 | Delhi | Kolkata | 40 | 40 | 6 | ₹13,357 | ₹2,482 | ₹53,209 | 14.38 | 34 |
| RT4 | Delhi | Hyderabad | 40 | 40 | 6 | ₹12,376 | ₹2,050 | ₹61,889 | 12.65 | 33 |
| RT5 | Delhi | Chennai | 40 | 40 | 6 | ₹19,603 | ₹1,998 | ₹70,323 | 15.80 | 29 |
| RT6 | Mumbai | Delhi | 40 | 40 | 6 | ₹14,809 | ₹2,336 | ₹59,964 | 10.55 | 32 |
| RT7 | Mumbai | Bengaluru | 40 | 40 | 5 | ₹12,375 | ₹2,074 | ₹74,880 | 13.41 | 37 |
| RT8 | Mumbai | Kolkata | 40 | 40 | 5 | ₹12,740 | ₹2,835 | ₹64,285 | 14.78 | 36 |
| RT9 | Mumbai | Hyderabad | 40 | 40 | 5 | ₹21,093 | ₹2,111 | ₹60,365 | 16.16 | 26 |
| RT10 | Mumbai | Chennai | 40 | 40 | 6 | ₹17,123 | ₹1,890 | ₹69,885 | 15.98 | 33 |
| RT11 | Bengaluru | Delhi | 40 | 40 | 6 | ₹11,996 | ₹2,723 | ₹66,411 | 11.72 | 34 |
| RT12 | Bengaluru | Mumbai | 40 | 40 | 5 | ₹19,003 | ₹3,012 | ₹83,916 | 13.83 | 31 |
| RT13 | Bengaluru | Kolkata | 40 | 40 | 6 | ₹16,061 | ₹3,393 | ₹73,388 | 13.79 | 32 |
| RT14 | Bengaluru | Hyderabad | 40 | 40 | 6 | ₹16,990 | ₹1,694 | ₹59,253 | 15.94 | 31 |
| RT15 | Bengaluru | Chennai | 40 | 40 | 6 | ₹14,582 | ₹1,604 | ₹65,445 | 18.54 | 34 |
| RT16 | Kolkata | Delhi | 40 | 40 | 5 | ₹18,094 | ₹3,999 | ₹76,255 | 14.76 | 29 |
| RT17 | Kolkata | Mumbai | 40 | 40 | 6 | ₹19,957 | ₹3,379 | ₹1,10,936 | 16.57 | 30 |
| RT18 | Kolkata | Bengaluru | 40 | 40 | 6 | ₹12,438 | ₹3,465 | ₹87,746 | 13.96 | 38 |

```
=AVERAGEIF(Bookings!C:C,A5,Bookings!K:K)
```

| Route_Id | Source_City | Destination_City | Total_Bookings | Unique_Flights | Airlines_Serving | Avg_Price | Min_Price | Max_Price | Avg_Duration_Hrs | Economy_Bookings |
|---|---|---|---|---|---|---|---|---|---|---|
| RT1 | Delhi | Mumbai | 40 | 40 | 5 | ₹15,969 | ₹2,381 | ₹71,056 | 10.91 | 30 |
| RT2 | Delhi | Bengaluru | 40 | 40 | 6 | ₹13,093 | ₹3,090 | ₹54,777 | 11.64 | 34 |
| RT3 | Delhi | Kolkata | 40 | 40 | 6 | ₹13,357 | ₹2,482 | ₹53,209 | 14.38 | 34 |
| RT4 | Delhi | Hyderabad | 40 | 40 | 6 | ₹12,376 | ₹2,050 | ₹61,889 | 12.65 | 33 |
| RT5 | Delhi | Chennai | 40 | 40 | 6 | ₹19,603 | ₹1,998 | ₹70,323 | 15.80 | 29 |
| RT6 | Mumbai | Delhi | 40 | 40 | 6 | ₹14,809 | ₹2,336 | ₹59,964 | 10.55 | 32 |
| RT7 | Mumbai | Bengaluru | 40 | 40 | 5 | ₹12,375 | ₹2,074 | ₹74,880 | 13.41 | 37 |
| RT8 | Mumbai | Kolkata | 40 | 40 | 5 | ₹12,740 | ₹2,835 | ₹64,285 | 14.78 | 36 |
| RT9 | Mumbai | Hyderabad | 40 | 40 | 5 | ₹21,093 | ₹2,111 | ₹60,365 | 16.16 | 26 |
| RT10 | Mumbai | Chennai | 40 | 40 | 6 | ₹17,123 | ₹1,890 | ₹69,885 | 15.98 | 33 |
| RT11 | Bengaluru | Delhi | 40 | 40 | 6 | ₹11,996 | ₹2,723 | ₹66,411 | 11.72 | 34 |
| RT12 | Bengaluru | Mumbai | 40 | 40 | 5 | ₹19,003 | ₹3,012 | ₹83,916 | 13.83 | 31 |
| RT13 | Bengaluru | Kolkata | 40 | 40 | 6 | ₹16,061 | ₹3,393 | ₹73,388 | 13.79 | 32 |
| RT14 | Bengaluru | Hyderabad | 40 | 40 | 6 | ₹16,990 | ₹1,694 | ₹59,253 | 15.94 | 31 |
| RT15 | Bengaluru | Chennai | 40 | 40 | 6 | ₹14,582 | ₹1,604 | ₹65,445 | 18.54 | 34 |
| RT16 | Kolkata | Delhi | 40 | 40 | 5 | ₹18,094 | ₹3,999 | ₹76,255 | 14.76 | 29 |

```
=COUNTIFS(Bookings!F:F,"Economy",Bookings!C:C,A6)
```

| Source_City | Destination_City | Total_Bookings | Unique_Flights | Airlines_Serving | Avg_Price | Min_Price | Max_Price | Avg_Duration_Hrs | Economy_Bookings | Business_Bookings |
|---|---|---|---|---|---|---|---|---|---|---|
| Delhi | Mumbai | 40 | 40 | 5 | ₹15,969 | ₹2,381 | ₹71,056 | 10.91 | 30 | 10 |
| Delhi | Bengaluru | 40 | 40 | 6 | ₹13,093 | ₹3,090 | ₹54,777 | 11.64 | 34 | 6 |
| Delhi | Kolkata | 40 | 40 | 6 | ₹13,357 | ₹2,482 | ₹53,209 | 14.38 | 34 | 6 |
| Delhi | Hyderabad | 40 | 40 | 6 | ₹12,376 | ₹2,050 | ₹61,889 | 12.65 | 33 | 7 |
| Delhi | Chennai | 40 | 40 | 6 | ₹19,603 | ₹1,998 | ₹70,323 | 15.80 | 29 | 11 |
| Mumbai | Delhi | 40 | 40 | 6 | ₹14,809 | ₹2,336 | ₹59,964 | 10.55 | 32 | 8 |
| Mumbai | Bengaluru | 40 | 40 | 5 | ₹12,375 | ₹2,074 | ₹74,880 | 13.41 | 37 | 3 |
| Mumbai | Kolkata | 40 | 40 | 5 | ₹12,740 | ₹2,835 | ₹64,285 | 14.78 | 36 | 4 |
| Mumbai | Hyderabad | 40 | 40 | 5 | ₹21,093 | ₹2,111 | ₹60,365 | 16.16 | 26 | 14 |
| Mumbai | Chennai | 40 | 40 | 6 | ₹17,123 | ₹1,890 | ₹69,885 | 15.98 | 33 | 7 |
| Bengaluru | Delhi | 40 | 40 | 6 | ₹11,996 | ₹2,723 | ₹66,411 | 11.72 | 34 | 6 |
| Bengaluru | Mumbai | 40 | 40 | 5 | ₹19,003 | ₹3,012 | ₹83,916 | 13.83 | 31 | 9 |
| Bengaluru | Kolkata | 40 | 40 | 6 | ₹16,061 | ₹3,393 | ₹73,388 | 13.79 | 32 | 8 |
| Bengaluru | Hyderabad | 40 | 40 | 6 | ₹16,990 | ₹1,694 | ₹59,253 | 15.94 | 31 | 9 |
| Bengaluru | Chennai | 40 | 40 | 6 | ₹14,582 | ₹1,604 | ₹65,445 | 18.54 | 34 | 6 |
| Kolkata | Delhi | 40 | 40 | 5 | ₹18,094 | ₹3,999 | ₹76,255 | 14.76 | 29 | 11 |
| Kolkata | Mumbai | 40 | 40 | 6 | ₹19,957 | ₹3,379 | ₹1,10,936 | 16.57 | 30 | 10 |
| Kolkata | Bengaluru | 40 | 40 | 6 | ₹12,438 | ₹3,465 | ₹87,746 | 13.96 | 38 | 2 |

## 5. ANALYSIS

- Combined information from different tables using lookup functions.
- Used **VLOOKUP and XLOOKUP** to retrieve required information.
- Created a **Route** column using Source City and Destination City.
- Created **Booking Window** based on Days Left.
- Created **Duration Category** based on flight duration.

- Created **Price Category** based on ticket price.
- Prepared the final dataset for analysis.

=CONCATENATE(G7," -> ",H7)

| G | H | I | J | K | L | M | N | O | P |
|---|---|---|---|---|---|---|---|---|---|
| ce_City | Destination_City | Stops | Duration_Hrs | Duration_Category | Price | Price_Category | Days_Left | Booking_Window | Routes |
| i | Mumbai | 0 | 2.17 | Short | ₹5,956 | Medium | 14 | Short Notice | Delhi -> Mumbai |
| i | Mumbai | 0 | 2.08 | Short | ₹28,872 | High | 15 | Short Notice | Delhi -> Mumbai |
| i | Mumbai | 0 | 2.25 | Short | ₹2,410 | Low | 30 | Moderate | Delhi -> Mumbai |
| i | Mumbai | 1 | 8 | Medium | ₹7,907 | Medium | 38 | Advance | Delhi -> Mumbai |
| i | Mumbai | 1 | 26.42 | Long | ₹41,281 | High | 19 | Moderate | Delhi -> Mumbai |
| i | Mumbai | 1 | 13.83 | Long | ₹70,793 | High | 0 | Last Minute | Delhi -> Mumbai |
| i | Mumbai | 1 | 24.42 | Long | ₹17,348 | High | 2 | Last Minute | Delhi -> Mumbai |
| i | Mumbai | 1 | 26 | Long | ₹15,353 | Medium | 2 | Last Minute | Delhi -> Mumbai |

M7

=IF(L7="","",IF(L7<=4631,"Low",IF(L7<=15800,"Medium","High")))

| G | H | I | J | K | L | M | N | O | P | Q |
|---|---|---|---|---|---|---|---|---|---|---|
| Source_City | Destination_City | Stops | Duration_Hrs | Duration_Category | Price | Price_Category | Days_Left | Booking_Window | Routes | |
| Delhi | Mumbai | 0 | 2.17 | Short | ₹5,956 | Medium | 14 | Short Notice | Delhi -> Mumbai | |
| Delhi | Mumbai | 0 | 2.08 | Short | ₹28,872 | High | 15 | Short Notice | Delhi -> Mumbai | |
| Delhi | Mumbai | 0 | 2.25 | Short | ₹2,410 | Low | 30 | Moderate | Delhi -> Mumbai | |
| Delhi | Mumbai | 1 | 8 | Medium | ₹7,907 | Medium | 38 | Advance | Delhi -> Mumbai | |
| Delhi | Mumbai | 1 | 26.42 | Long | ₹41,281 | High | 19 | Moderate | Delhi -> Mumbai | |
| Delhi | Mumbai | 1 | 13.83 | Long | ₹70,793 | High | 0 | Last Minute | Delhi -> Mumbai | |
| Delhi | Mumbai | 1 | 24.42 | Long | ₹17,348 | High | 2 | Last Minute | Delhi -> Mumbai | |
| Delhi | Mumbai | 1 | 26 | Long | ₹15,353 | Medium | 2 | Last Minute | Delhi -> Mumbai | |

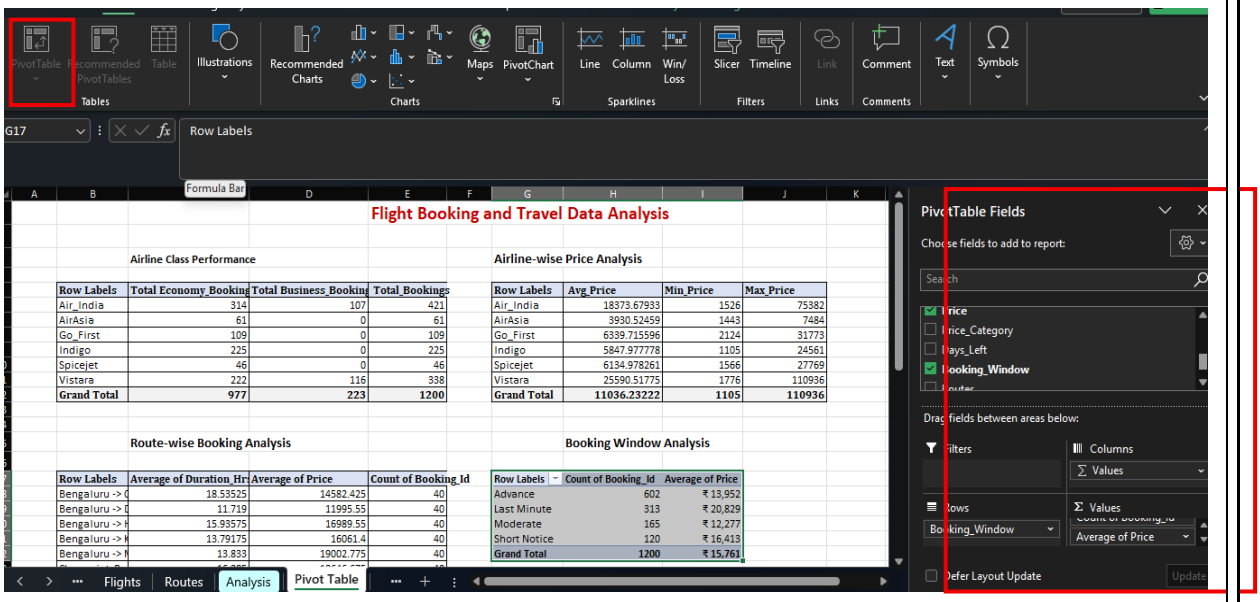
H7

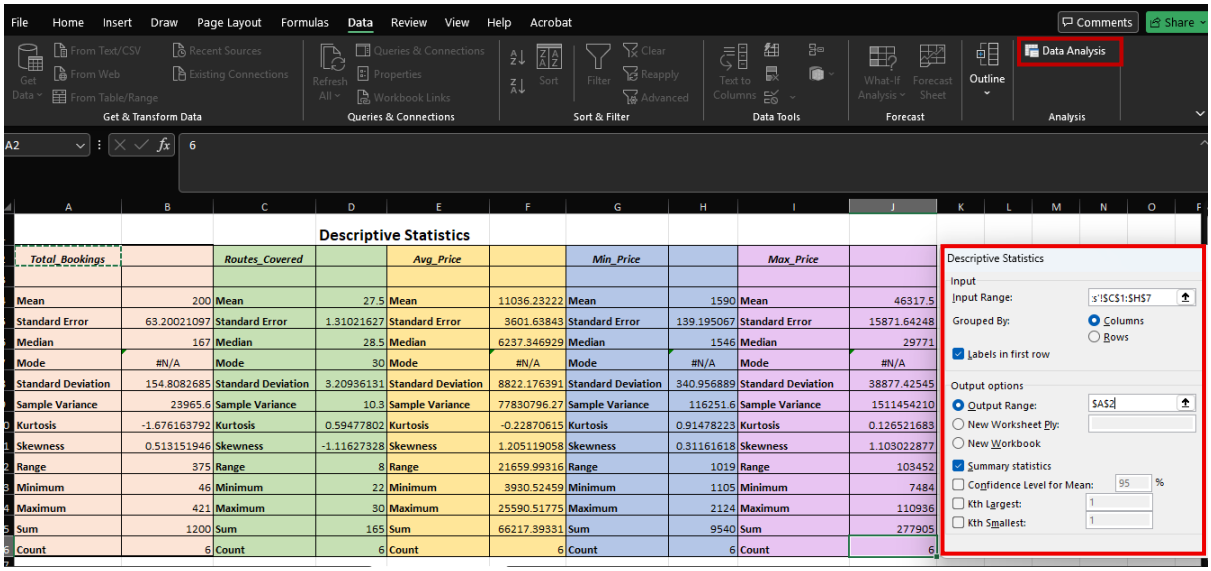
=VLOOKUP(B7,Flights!A:H,6,FALSE())

| A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|
| Booking_Id | Flight_Id | Route_Id | Airline_Id | Airline_Name | Class | Source_City | Destination_City | Stops | Duration_Hrs | Duration_Category |
| BK1000 | FLT10002 | RT1 | AL100 | AirAsia | Economy | Delhi | Mumbai | 0 | 2.17 | Short |
| BK1001 | FLT10006 | RT1 | AL105 | Vistara | Business | Delhi | Mumbai | 0 | 2.08 | Short |
| BK1002 | FLT10010 | RT1 | AL102 | Go_First | Economy | Delhi | Mumbai | 0 | 2.25 | Short |
| BK1003 | FLT10030 | RT1 | AL102 | Go_First | Economy | Delhi | Mumbai | 1 | 8 | Medium |
| BK1004 | FLT10070 | RT1 | AL105 | Vistara | Business | Delhi | Mumbai | 1 | 26.42 | Long |
| BK1005 | FLT10073 | RT1 | AL105 | Vistara | Business | Delhi | Mumbai | 1 | 13.83 | Long |
| BK1006 | FLT10077 | RT1 | AL105 | Vistara | Economy | Delhi | Mumbai | 1 | 24.42 | Long |
| BK1007 | FLT10078 | RT1 | AL105 | Vistara | Economy | Delhi | Mumbai | 1 | 26 | Long |
| BK1008 | FLT10079 | RT1 | AL103 | Indigo | Economy | Delhi | Mumbai | 1 | 4.17 | Medium |
| BK1009 | FLT10081 | RT1 | AL103 | Indigo | Economy | Delhi | Mumbai | 1 | 5.08 | Medium |

## 6. PIVOT TABLE:

- Created PivotTables to summarize the cleaned data.
- Performed airline-wise booking analysis.
- Performed class-wise booking analysis.
- Analyzed average, minimum and maximum ticket prices.
- Compared booking and pricing information across airlines.




## 7. DESCRIPTIVE STATISTICS:

- Calculated descriptive statistics for the required numerical variables.
- Calculated Mean, Median, Mode
- Calculated Standard Deviation, Variance, Range.
- Calculated Minimum and Maximum.
- Calculated Sum and Count.
- Calculated Skewness and Kurtosis, Standard Error.